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StepIdeator: Utilizing Mixed Representations to Support Step-By-Step Design With Generative Artificial Intelligence

In the product conceptual design, designers utilize multiple design representations to ideate, externalize, and refine concepts iteratively. Mixed representations, defined as the simultaneous presentation of multiple representations, foster deeper insights and facilitate broader exploration compared to focusing on a single representation. However, designers often struggle with the cumbersome process of creating, transforming, and refining these representations. Advanced artificial intelligence (AI) capabilities now significantly lower the barriers to creating text, images, and 3D models, presenting substantial potential for application in design practice. However, the current design tools based on generative AI tend to produce polished product images directly, hindering designers from continuously deliberating on mixed representations and thus limiting their creative potential. Therefore, on the basis of a formative study, we propose StepIdeator, a step-by-step design tool driven by generative AI. This tool facilitates seamless transition and refinement of mixed-design representations. Through a comparative study (N = 16), we validated StepIdeator's effectiveness in improving idea externalization and enhancing creativity. Furthermore, the results revealed that designers perceived greater confidence, ownership, and sense of contribution when collaborating with StepIdeator. [DOI: 10.1115/1.4067426]

Keywords: artificial intelligence, conceptual design, creativity and concept generation

1 Introduction

During the conceptual design process, designers utilize various design representations to externalize and refine design concepts [1,2]. Design representations, including texts, sketches, 3D models, and product renderings [3,4], aid in translating abstract ideas into concrete schemes and improve communication among designers, engineers, clients, and stakeholders [5,6]. Each representation offers distinct insights but meanwhile exhibits limitations. For instance, sketches enable rapid conceptualization but lack details [7], while 3D models offer comprehensive details but are time intensive to produce. Therefore, in the design process, designers always alternate or cycle through different representations to gain comprehensive insights, analyze problems, and produce unique ideas [8,9].

The evolution of computer-aided design tools has transformed the medium of design representations, enabling designers to create in both physical and digital spaces. In the early stages of design, designers define abstract concepts with tools such as documents and mind maps, translating design challenges and client requirements into specifications through natural language. Designers then sketch these concepts using traditional paper or digital

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Contributed by the Design Automation Committee of ASME for publication in the Journal of Mechanical Design. Manuscript received April 30, 2024; final manuscript received December 2, 2024; published online January 30, 2025. Assoc. Editor: Zhenghui Sha.

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tools like PHOTOSHOP[2] and SKETCHBOOK.[3] Parametric modeling software such as RHINO[4], SOLIDWORKS[5], and AUTOCAD[6] is utilized for model development, reuse, and alteration. Additionally, rendering software like KEYSHOT[7] and BLENDER[8] allows the visualization of the color, materials, and finishes (CMF) of products through camera, material, and lighting settings. To conclude, designers switch between various tools and mediums to progressively express their ideas through different design representations.

Compared to using a single design representation simultaneously, incorporating multiple representations (e.g., sketches, models) into a unified design environment has been proven to enhance creativity, efficiency, and innovation significantly [10]. This mixed-representation environment enables designers to seamlessly navigate between different representations and to dynamically adjust their working focus in response to emerging insights [11]. Such flexibility promotes in-depth design exploration and enhances decision-making processes, yielding innovative solutions [12]. However, most designers still utilize different representations as separate parts in practice. The widespread adoption of a unified design environment remains limited by several challenges. First, the wide variety of creation methods and tools for design representations complicates transitions between different media and dimensions [13], resulting in a lack of tools that support both flexible switching and simultaneous presentation of multiple representations for designers. Second, the most essential design representations to designers and the specific challenges of utilizing them remain unclear.

As generative artificial intelligence (AI) gains the capability to generate various representations rapidly, AI-based tools have been prevalently used in design activities to help designers cope with the aforementioned challenges. Tools such as CHATGPT[9], MIDJOURNEY[10], and STABLE DIFFUSION[11] can interpret textual descriptions from designers, make appropriate suggestions, and generate vivid images. Leveraging these abilities, researchers have developed creative support tools (CSTs) to automate concept creation, optimize product details, and enable rapid exploration of a wide range of design alternatives [14]. For example, tools like PAINTER [15] adapt large language models (LLMs) to realize controllable text-to-sketch generation, supporting the refinement of sketch details through text modifications. Researchers also enhance the quality of generation during the transition between two representations by introducing a third medium, such as sketch-guided text-to-3D tools [16,17]. Additionally, Edwards et al.'s framework facilitates the rapid generation of text, images, and 3D models across sketches, enhancing early-stage design exploration [18].

The aforementioned studies are primarily intended to help designers achieve polished final representations, yet the purpose of design goes beyond by merely producing aesthetically pleasing outputs [19]. Conceptual design entails a thoughtful process of synthesizing solutions within defined constraints, involving collaborative evaluation of requirements, functionality, and logic [1,9]. This progression naturally aligns with the design process of externalizing thoughts through step-by-step engagement with different representations [5]. Current research and tools based on generative AI do not align with the conceptual design process for two main reasons: (1) AI-based CSTs tend to adapt design processes to AI rather than tailoring AI to designers' workflows, resulting in an "AI-centric" approach [20] that is inconsistent with designers' actual working habits. For instance, researchers focus mainly on the capability of modal transformation, including design retrieval and recommendation [21–23], representation generation [24–26], and interrepresentation manipulation [27]. (2) AI's one-step generation reduces designers' reflection time, risking premature convergence and design fixation unless designers are explicitly encouraged to generate diverse ideas from the start [28–30]. Thus, there is a pressing need for design representation tools that not only prioritize the conceptual design process but also seamlessly integrate AI to complement, rather than dictate designers' workflows.

To this end, our study focuses on exploring the potential of an AI-based mixed-representation environment to support design activities. The term "mixed" implies two aspects. First, it breaks down AI's generation process into a manageable, step-by-step generation of different design representations. Second, it supports designers to seamlessly switch and navigate between various combinations of representations. This research includes three phases. Initially, to grasp designers' preferences and strategies about multiple design representations, we conducted a formative study and identified four representations central to our research. We also pinpointed three key challenges when integrating generative AI for externalizing design thoughts. Second, we designed and implemented a mixed-representation system named StepIdeator. It was designed to provide targeted inspiration, externalize design ideas, and assist designers in flexibly switching between representations. Finally, in a comparative study involving 16 designers, we compared StepIdeator against a baseline system. The findings revealed that StepIdeator enhanced idea externalization and creativity. By analyzing the count of combinations and transformations of mixed-design representations, we explored designers' preferences for these combinations and identified three sources of inspiration within the transformations. Participants also reported a heightened sense of ownership and confidence, indicating a substantial improvement in the human–AI collaboration experience.

This article makes three main contributions.

(1) Revealing designers' challenges when employing generative AI for externalizing design thoughts, and proposing design goals (DGs) for developing mixed-representation tools through the formative study.
(2) Developing StepIdeator, an AI-based system that supports the iterative development and refinement of design concepts through appropriate representations.
(3) Validating the effectiveness of StepIdeator and discussing how the combinations and transitions of representations influence creativity and authorship.

2 Related Works

2.1 Representations in Conceptual Design. Conceptual design is the foundational stage of product design, where ambiguous design issues are transformed into concrete design concepts and prototypes [9,31]. It is an intricate endeavor that demands creativity, expert insight, experience, and also discernment in relation to both the practical and artistic facets of a specific field [19]. In this process, designers need to explore the design space as much as possible to increase the product's chances of success in the market [32–34].

Design representation is crucial for enhancing designers' learning, analysis, communication, conceptualization, cognition, and the redesign of design artifacts [35]. It involves methods for abstracting and graphically depicting design information, knowledge, and the relationships and structure of notional objects. Designers primarily employ various representations during the conceptual design process, including design briefs (DBs), sketches, detailed drawings, models, product renderings, and scenarios [4,13,36]. Specifically, design briefs consist of linguistic content detailing customer requirements, design rules, constraints, and verbal communications between stakeholders [37]. Sketches are basic visuals of an object with essential outlines and may include dimensions, stroke variations, and shading, serving as a base for

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[3]https://www.sketchbook.com/
[4]https://www.rhino3d.com/
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detailed designs [7,8]. In detailed drawings, designers use markers and colored pencils to create realistic products with full color [4]. Models refer to three-dimensional visual representations used to reproduce the physical attributes of an intended product [38]. Renderings, including product renderings and scenarios, show the CMF of product concepts vividly.

Each design representation not only offers unique insights but also has limitations [1]. For instance, sketches facilitate quick conceptualization but are deficient in details [7], whereas 3D models provide extensive details but require considerable time to create. Sole reliance on one form might neglect some design facets. Therefore, designers engage with diverse representations to align with the progressive workflows, which include initiating divergent thinking, collecting references, refining iterations, and validating concepts [9]. For instance, when creating sketches, designers typically collect references and roughly consider the layout [4]. During the modeling phase, they further consider the feasibility of the physical structure and optimize parting lines, thereby validating concepts [1]. This process facilitates the extraction of pivotal insights essential for innovatively addressing the intricacies of design [32]. To better support the ideation process, researchers have developed various methodologies to classify design representations, outline their respective strengths and weaknesses [4,5], and discuss their optimal use across the design process [9,39,40]. Through these methodologies, designers use the appropriate representations at the right time to facilitate design workflows. This allows the design idea to gradually develop from an abstract semantic concept to sketches and models, progressively advancing toward production. This process is iterative rather than linear, facilitating continuous refinement and optimization [41]. With each transition between representation forms, designers' thinking evolves, enhancing their understanding of design issues [42], fostering idea generation [1], and confirming decisions [40].

Further studies have shown that using a mix of sketches, models, and other design representations within a unified design environment boosts creativity, efficiency, and innovation [10]. Researchers have pinpointed three key types of switching behaviors beneficial in design environments, including shifting focus among different visual aspects and alternating between representations [12]. Although this approach aided in making appropriate design decisions and boosting co-evolution through successive iterations of representations, it also introduced the challenge of managing multiple tools and software to transform representations, complicating the design workflow. Therefore, tools are needed that support designers in flexibly switching between and simultaneously presenting multiple representations throughout the design process.

2.2 Creativity Support Tools for Representations. With the growing adoption of data science and AI in design processes, CSTs for representations are attracting more attention [43]. These tools focus on the modalities of text, sketches, and models to enhance product design by facilitating scientific analysis, fostering creative thinking, and aiding in problem-solving [13]. Some researchers employ deep learning techniques for solving cross-modal tasks to support different steps of conceptual design [13], while others utilize Augmented Reality (AR) and Virtual Reality (VR) to bridge the gap between digital and physical spaces in the ideation process [44,45].

To help designers collect references and inspire design ideas, some CSTs support retrieval and recommendation functions, searching existing design repositories of different representations. Some researchers focus on the interactions between two of the three modalities [13]. For example, Yang et al. proposed a sequential learning framework for sketch-based 3D model retrieval to learn 3D model representation and 2D sketch representation separately and sequentially [23]. Similarly, researchers explore text-to-3D shape retrieval [46] and further discuss the potential of sketch and text-to-3D shape retrieval in mixed reality [47]. As an example, SHAPEFINDAR used in situ sketches blended with textual queries to search for 3D models on HoloLens 2 [48].

To facilitate design space exploration and fast ideation, researchers have developed CSTs to support interrepresentation manipulation for designers to refine their ideas. For instance, Control3d interactively controlled and shaped the synthetic 3D contents according to users' sketches and texts to reduce modeling costs and skill requirements [17], while Clip-Nerf enabled designers to individually deform the shape or change the appearance color of 3D objects based on a text prompt or a reference image [49]. To streamline fast modeling, researchers also explore conceptual modeling in VR, linking 2D sketches and 3D models in a unified environment [44,45,50].

Despite these advancements, applying existing CSTs to conceptual design poses challenges. Designers often gather extensive information across various design steps to inspire their creations, yet existing CSTs typically focus on a single specific task at a time. This narrow focus hinders swift and adaptable transitions among various design representations based on designers' changing needs. Moreover, these systems may lack the capability to synthesize and generate new ideas from real-time design data provided by designers, limiting their utility in the early stages of design where creative flexibility is paramount.

2.3 Generative Artificial Intelligence for Conceptual Design. Generative AI can create seemingly new texts, images, audio, and other media based on existing data and algorithms [51]. The widespread adoption of this technology with examples such as CHATGPT and STABLE DIFFUSION is currently revolutionizing designers' workflows.

First, as designers find difficulty in articulating abstract design concepts in texts [52] or further transforming semantics into visual images [53], researchers leverage AI's rapid and cross-modal generative capabilities to directly generate and refine design solutions. For example, some AI-assisted tools have automated specific design processes: STYLEME generates designer-style clothing sketches and provides intelligent colorization [54], while BIGNET focuses on extracting brand-related features from product images [55]. Other AI-assisted tools like PAINTER [15] adapted LLMs to support text-to-sketch generation. Researchers have also developed algorithms to generate 3D meshes from freehand sketches, aiming at fast 3D modeling [24,56]. Similarly, Jain et al. combined neural rendering with multimodal images and texts to synthesize diverse 3D objects solely from natural language descriptions [25]. The effectiveness of generating design solutions from concepts is further enhanced by incorporating a third form, like using sketches to guide text-to-3D conversion tools [16,17].

Second, to mitigate design fixation and enhance creativity, tools such as POPBLENDS [57] and GANSPIRATION [58] used generative AI to produce creative stimuli across various modalities, including texts and images. Systems including SEMANTICCOLLAGE [59] and GANCOLLAGE [60] added digital mood boards to semantic labels to facilitate divergent thinking. Designers used tools such as MOODCUBES [61] and METAMAP [62] to further investigate design materials across various dimensions. Researchers have also utilized AI to enhance the transferability of design knowledge [63] and parameters [64] and have employed spoken annotation and design operations to capture and frame design intentions [65]. Consequently, generative AI has been proven to enhance human creativity by recognizing complex patterns, generating new content, and collaborating with humans [20,66,67].

However, current research and tools based on generative AI do not align with the conceptual design process for two main reasons. First, previous studies have focused more on "designing for AI, not designing with AI" [20], leading to approaches inconsistent with designers' actual working habits. Specifically, the ambiguity in designers' concepts clashes with the clear inputs required by AI [52], complicating coordination. Additionally, existing tools emphasize generating polished outcomes instead of supporting a collaborative assessment of needs, functions, and logic [1,9]. Second, AI's ability to generate designs in one step reduces

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designers' reflection time, risking premature convergence and design fixation unless designers are explicitly encouraged to generate diverse ideas from the start [28–30]. This issue particularly challenges novice designers, who often quickly enter the solution space with minimal problem space exploration [68].

In conclusion, ideal design tools should prioritize the conceptual process and complement designer workflows rather than dictate them. Therefore, further study is needed to understand how designers select and cycle through design representations and their challenges with generative AI in externalizing design concepts.

3 Formative Study

Considering the existing research gap, we conducted a formative study with eight designers to grasp designers' preferences and strategies for multiple design representations and to understand how AI can support designers in creating representations. Based on previous research studies [4,5,9], the study focused on five common representations: design briefs, sketches, detailed drawings, 3D models, and renderings (Fig. 1).

3.1 Participants. We recruited eight product designers (D1–D8, age: $M = 23.25$, $SD = 1.34$) via social media, each with at least 3 years of experience. All designers were familiar with generative AI, such as CHATGPT, MIDJOURNEY, and STABLE DIFFUSION, with six having applied AI to the product design. Designers signed an informed consent form and received monetary compensation after completing the study.

3.2 Procedure. *Semistructured interviews (15 min)* were conducted to understand designers' strategies and preferences regarding design representations. Responses were analyzed to document the frequency of each representation's use and examine the routine integration of generative AI in conceptual design.

Design representation generation tasks (60 min) were conducted to identify designers' challenges with AI in externalizing design concepts. The tasks were divided into five 12-min sessions. After reviewing definitions and examples of a specific representation, designers first engaged in a 10-min generation task for each type of design representation: GPT for design briefs; STABLE DIFFUSION for sketches, flat model images, and renderings; and SUDOAI [69] for quick generation of editable model formats. These models were chosen for their demonstrated state-of-the-art capabilities in generating high-quality outputs of their respective representations. Immediately following each task, designers spent approximately 2 min filling out a questionnaire assessing the pros and cons on a seven-point scale, informed by the prior research studies [4,9,32,39,40]. They were also encouraged to add any additional benefits or challenges they encountered. Then designers evaluated their satisfaction with AI-generated outcomes using seven-point scales.

Postsurvey interviews (15 min) were conducted to understand designers' expectations of AI assistance in utilizing design representations during the design process. The researchers explained the purpose of the study and our preliminary solutions to the designers, soliciting their feedback as a basis for further system design. All the interview questions in semistructured interviews and postsurvey interviews are listed in Appendix A.



Fig. 1 Examples of five design representations discussed in the formative study, including design briefs (text format), sketches (image format), detailed drawings (image format), 3D models (image and OBJ format), and renderings (image format)

3.3 Finding 1: Changes in the Creation of Different Representations After Integrating Artificial Intelligence. We extracted the characteristics of creating design representations using traditional methods from previous research studies [4,7,9,12]. These characteristics were then distilled into benefits and challenges, covering dimensions such as characteristics of medium, functionality, usability, the degree of idea externalization, and the ease of switching between representations (as outlined in Fig. 2). By incorporating designers' subjective ratings of these benefits and challenges during the experiment, we aim to explore how generative AI can help designers maximize these benefits and address existing challenges. Naturally, integrating AI may also introduce new issues not covered in the previous research, which are supplemented by insights from interviews. We will discuss all of these topics as follows.

3.3.1 Design Briefs. Tools like GPT not only improved the efficiency of organizing design briefs (DB-B1, DB-B3, and DB-B4) but also enriched content comprehensively (DB-B2). AI's multimodal generative capabilities help designers convert design briefs into visual concepts (DB-C1). However, designers evaluated that AI-generated design briefs primarily offer generic information about existing products, "lacking customization for specific design contexts." Furthermore, these briefs often "fail to thoroughly analyze the designers' pain points, resulting in a mismatch with the designers' actual needs."

3.3.2 Sketches. AI-generated sketches offer structural reference value (S-B1, S-B2) and can quickly produce multiple styles (S-C3), benefiting designers exploring different design concepts in the early stages. However, these sketches might contain impractical details, with "functions sometimes not matching the specific structures," necessitating numerous later adjustments by designers, which "fails to significantly reduce the workload." Also, there is a lack of innovation in appearance, "showing little difference from existing products" and "a lack of a deductive process for form creation." Moreover, to clearly present designs, "more annotations and views are often needed." AI struggles to truly understand hints related to style and tone, further limiting its capability in design innovation.

3.3.3 Detailed Drawings. Interestingly, detailed drawings' functionality may be diminished. AI-generated vivid renderings can replace detailed drawings for depicting CMF (DD-B1), overcoming limitations related to the designer's drawing time and technique (DD-C2). For example, D5 believes that AI can quickly generate CMF schemes and apply them to models for selection; D1, D3, and D4 "rarely use detailed drawings, typically experimenting with CMF during rendering and making repeated adjustments." Similarly, the function of detailed drawings in distinguishing structures (DD-B2) can also be replaced by AI-generated models. For instance, D2, D5, and D7 suggest that if AI can generate model images or 3D reconstructions directly, compared to detailed drawings, it would "better facilitate structural analysis and timely design modifications."

3.3.4 3D Models. AI offers two model generation formats: a series of flat images or 3D reconstructions. These rapid and low-cost model visuals help designers choose the optimal design schemes before modeling, reducing ineffective trials and adjustments during the design process (M-C1, M-C2, and M-C3). AI-provided models offer crucial references for creating partition lines or structural lines, aiding in the layout and arrangement of internal components. Nonetheless, using AI-generated model visuals also faces challenges such as "models lacking functional

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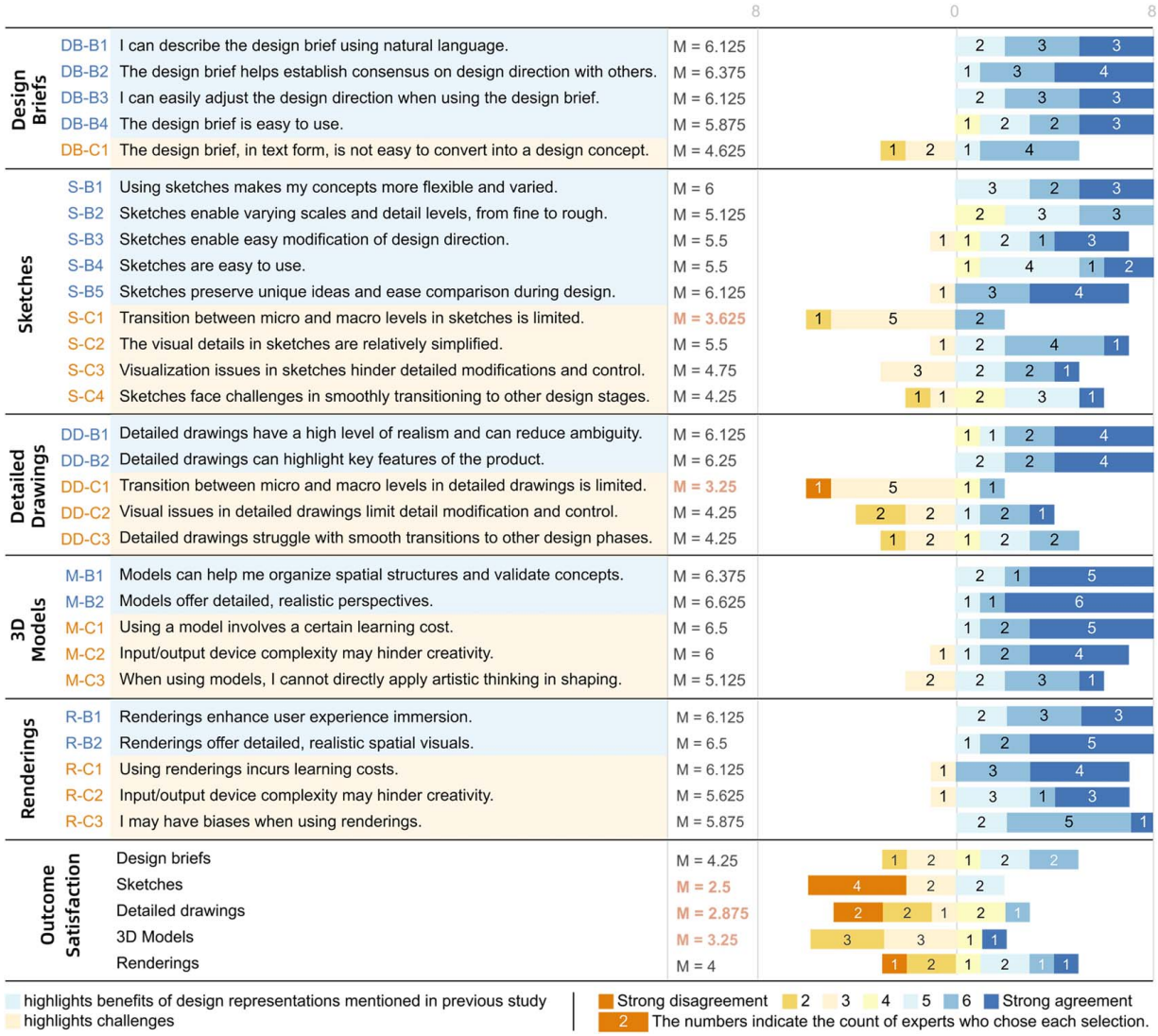


Fig. 2 Evaluation results of the benefits and challenges of using five different representations

correlation," "inability to clearly demonstrate the product's structural performance," and "unreasonable proportions and structures." Designers also note that the generated model visuals often do not align with their design intent, preferring to "generate models strictly based on own sketches."

For model generation, we sought designers' opinions in choosing between a series of multiangle model images generated by STABLE DIFFUSION and editable model files. Although the models generated by sudoAI offered better editability, designers found that the multiangle 2D model images were generated more quickly and "better met the needs of rapid iteration and verification required during the conceptual design phase." Additionally, the precision of the 2D model images was higher than that of the models generated by sudoAI, "providing greater assistance in refining structural details." Given the limitations in speed and quality of current generation models, we selected 2D images as the format for 3D models in the subsequent system design.

3.3.5 Renderings, Including Product Renderings and Scenarios. Generative AI allows designers to obtain scenarios in the early stages of design (R-C1), promoting divergent thinking and thereby inspiring more innovative design concepts. However, designers thought that AI-generated scenes often fail to accurately highlight key points. The presentation of product details is also frequently unsatisfactory, which may lead to results that do not align with the original design intent. Furthermore, AI in the design process often references existing products on the market, leading to similar appearance styling and coloring.

Overall, current general-purpose tools based on AI are not specifically tailored for design representations in product design, leading to significant time and effort spent correcting their outputs. For example, D2 used STABLE DIFFUSION to transform her sketches into model images for a feasible structural design of a vacuum cleaner. However, even to achieve a design representation that met conceptual design specifications, D2 invested substantial time in selecting the model and prompts. Worse still, by the end of the session, her design concept had not progressed beyond the initial idea. The designers' feedback on the AI-generated representations mentioned earlier also supports this point.

3.4 Finding 2: Changes in the Overall Design Workflow After Integrating Artificial Intelligence. As shown in Fig. 3, AI's intervention has greatly transformed designers' routine workflows by rapidly generating polished and concrete concepts. These detailed outputs would typically take several hours with traditional tools but can be achieved by AI in seconds to minutes. Therefore, AI enables designers to explore multiple design options swiftly and significantly jump from abstract conceptual text directly to highly detailed renderings. For example, D2, D4,

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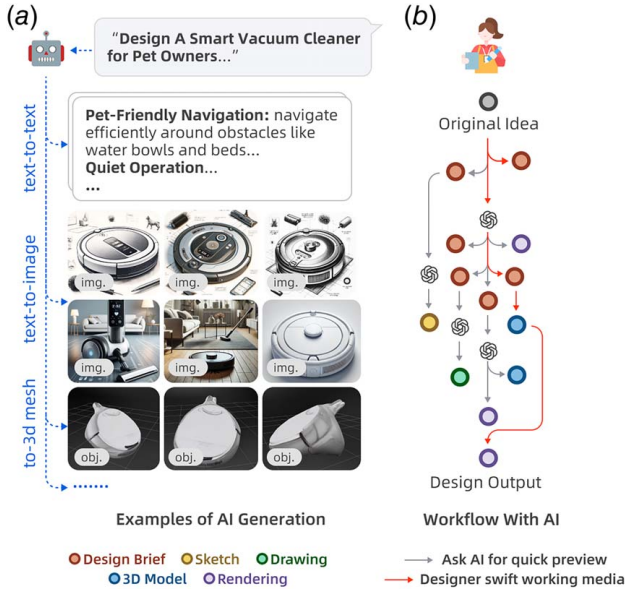


Fig. 3 Examples of design workflows after integrating AI: (*a*) AI's proficiency in generating textual, image, and 3D mesh outputs, demonstrating its potential to generate content approximating design representations. (*b*) How designers utilize AI and shift working representations and media in conceptual design is shown.

and D7 generated multiple product styling references. D7 also used AI-generated renderings as auxiliary presentations in early client communications. Despite the boundaries of design representations appearing blurred by AI's integration, designers still emphasized the irreplaceability of design briefs, sketches, models, and renderings in finalizing design schemes. Each was mentioned eight times, more than other forms like storyboards (5), detailed drawings (3), etc.

However, generative AI may make the design workflows unidirectional and linear [70], directly leading to detailed solutions and restricting the depth of designers' involvement and engagement. The limitations of AI in cross context may also hinder designers' creative process and reduce the time for deep reflection and critical thinking. For instance, when generating sketches, many designers spent an average of only 3.4 minutes of a 10-min generation task on conceptualizing ideas, with most of their time spent adjusting prompts and parameters. Yet these efforts to guide the AI did not result in satisfactory outcomes, as reflected by the relatively low satisfaction with the sketch results ($M = 2.5$ in Fig. 2). Several designers (D3, D4, D5, and D7) noted, "AI provided me with highly detailed product sketches right off the bat, but I needed divergent thinking about functions, structures, and styling." D5 mentioned, "The generated images need more annotations for me to understand the design intent. The representations shown are only results, lacking sufficient inspiration or support for a thorough conceptualization process."

3.5 Finding 3: Designers' Expectations for Artificial Intelligence When Creating Representations. Based on the previous study [71], we have classified the assistance provided by AI in creating representations during the conceptual design process into four categories as follows (Fig. 4):

- *Create new* refers to AI generating new ideas or schemes for designers. During interviews, designers expressed their desire for AI to identify missed aspects, quickly generate multiple design options, and produce realistic and detailed specific scenes.
- *Refine* involves evaluating and amending a contribution with similar inputs. For all representations, designers expected AI to "offer constructive advice and solutions based on an understanding of existing products."
- *Extend* represents AI's ability to expand or add to any collaborator's previous contributions. Several designers noted that AI excelled at "providing specific information at a practical level" and had "strong detail expansion capabilities," which accelerated the advancement of design schemes.
- *Transform* denotes AI's capability to assist designers in switching between different representations. With its text-to-image and image-to-image abilities, AI helped designers "quickly preview potential effects and draw inspiration."

Despite these clear expectations, interviews and observations highlighted challenges in conveying detailed instructions to AI and AI's subsequent struggles to adhere strictly to a designer's vision. To align AI collaboration with designers' expectations, designers' focus in practice has shifted from externalizing original concepts to understanding and guiding AI-generated designs, thereby increasing communication costs. D3 mentioned, "In early design stages, when my concepts are still vague, conveying specific product appearances to AI is difficult. However, AI functions effectively only with precise prompts, complicating its use." D6 also noted, "By the time I can provide detailed instructions, the design concept has begun to take shape, and I increasingly seek convergent advice from AI, such as feasible structures and shapes." However, due to the inherent black-box nature and randomness of the generation process [72], generative AI may not produce outputs that

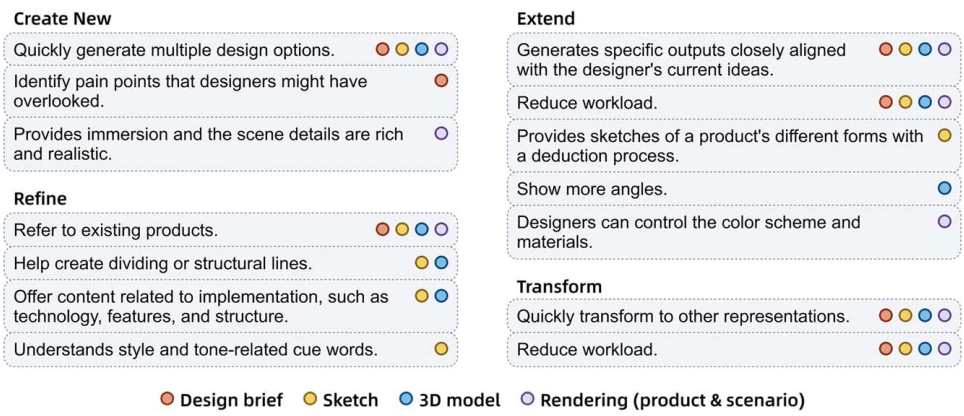


Fig. 4 Results of classifying designers' expectations for generative AI. This figure illustrates how designers expect AI to do in the conceptual design process: creating new ideas, refining existing concepts, extending designs, and transforming representations.

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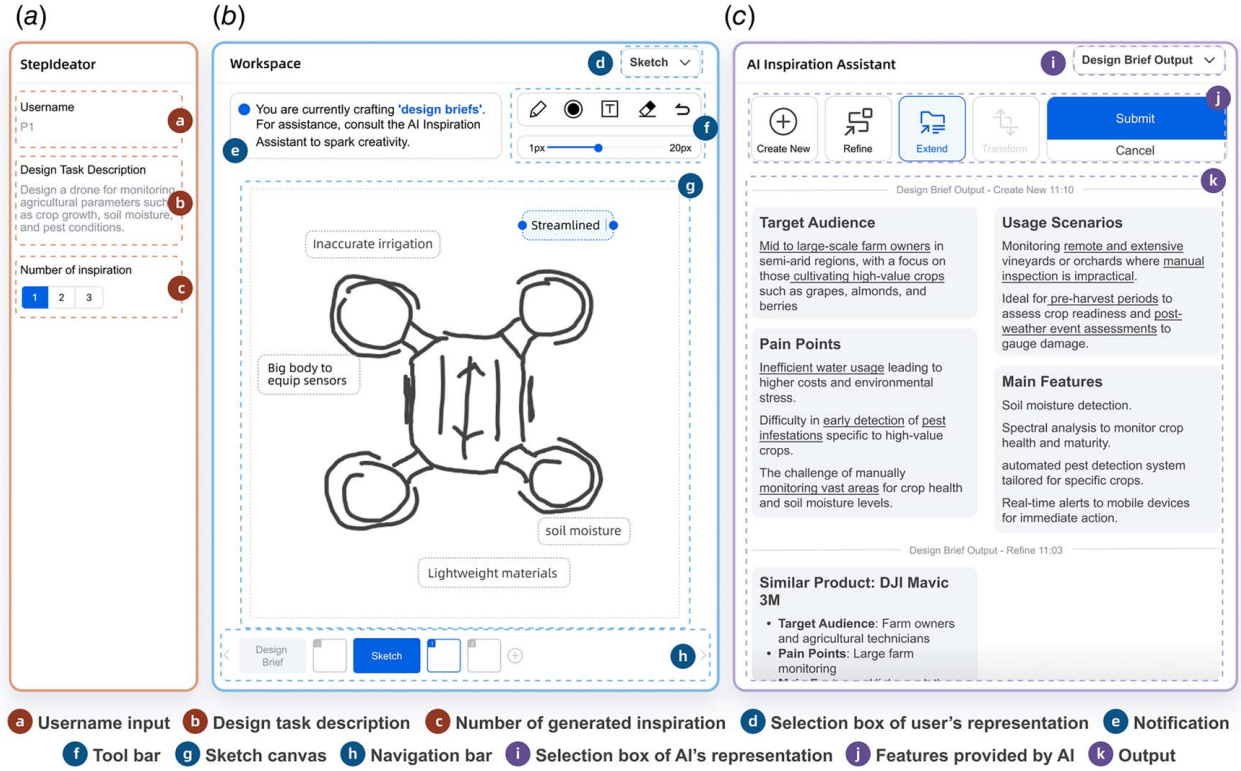


Fig. 5 The overview of StepIdeator. The main user interface of StepIdeator consists of three panels: (*a*) a Basic Information Panel, (*b*) a Sketch Canvas (Workspace), and (*c*) an AI Inspiration Assistant Panel. In both the Workspace and AI Inspiration Assistant Panel, designers have separate checkboxes to alter design representations. Within the Workspace, designers can toggle this option to record their ideas, while in the AI Inspiration Assistant Panel, they can modify it to adjust the representations outputted by the four features.

closely follow these instructions, especially with respect to the finer details of the design.

4 StepIdeator

4.1 Design Goals. We identified three DGs to build a system that iteratively supports designers' idea externalization during the conceptual design.

DG1: Precise Control Over Design Representation Creation for Better Expression. For free creative expression (finding 1), designers should have the autonomy to meticulously adjust and refine each representation through a step-by-step approach, either independently or with AI assistance.

DG2: AI-Driven Creativity Support Aligning With the Design Workflows. To provide ample time for iteration and reflection (finding 2), the system should deliver step-by-step features aligned with designers' workflow phases: sparking divergent thinking, simplifying reference gathering, facilitating iterations, and validating concepts.

DG3: Balance of Detailed Prompts and Ambiguous Conceptualization to Bridge the Human–AI Communication Gap. To alleviate the burden on designers of inputting detailed prompts (finding 3), the system should enable AI to capture users' intended ambiguity, vagueness, and imprecision.

4.2 System Overview. We implemented StepIdeator (Fig. 5), which supports mixed representation within an environment that includes design briefs, sketches, models, and renderings. Considering the stability of the outcomes, StepIdeator supports image formats for 3D models. Its renderings feature both detailed product visuals and situational scenarios. This system is designed to provide specific inspiration and facilitate effortless navigation among different representations.

StepIdeator integrates three panels for a streamlined design process. (1) The Basic Information Panel, capturing usernames and design tasks (Fig. 5(*a*)). (2) The "Workspace" Sketch Canvas is equipped with textual and drawing tools and allows users to add or switch between different canvases to document ideas, enabling the tracking and evolution of designs (Fig. 5(*b*)). (3) The AI Inspiration Assistant Panel shows AI-generated outputs and a suite of features (Fig. 5(*c*)).

For DG1, StepIdeator allows designers to modify and toggle independently between these representations in the Workspace and the AI Inspiration Assistant Panel. For example, designers can sketch in the Workspace while referencing models or renderings in the AI Inspiration Assistant Panel, or they can further modify AI-generated sketches placed in the Workspace. Changing the representation in the panel's selection box dictates the AI's output form, enhancing flexibility in the design process. Such a feature facilitates the easy shifting of focus and references, thereby enriching design creativity and productivity.

For DG2, the AI panel introduces four pivotal features—"Create New", "Refine", "Extend" and "Transform"-to generate corresponding outcomes and support progressive iteration. Each feature is aligned with a specific stage of the designer's workflow. Designers activate these features by clicking the corresponding button, followed by submitting inputs for AI processing. For example, as shown in Fig. 6, designers can utilize "Create New" to have AI propose new functions or create scenarios based on a design brief, enhancing early divergent thinking. If designers use "Extend" during sketching, AI assists by quickly refining drafts or providing other corresponding representations. Additionally, when "Transform" is employed, AI converts briefs or sketches into 3D models for swift previews of structures and forms. According to the selected representation in the Workspace and AI panel, we provide a total of 14 distinct functional operations to support interaction between the same and different representations.

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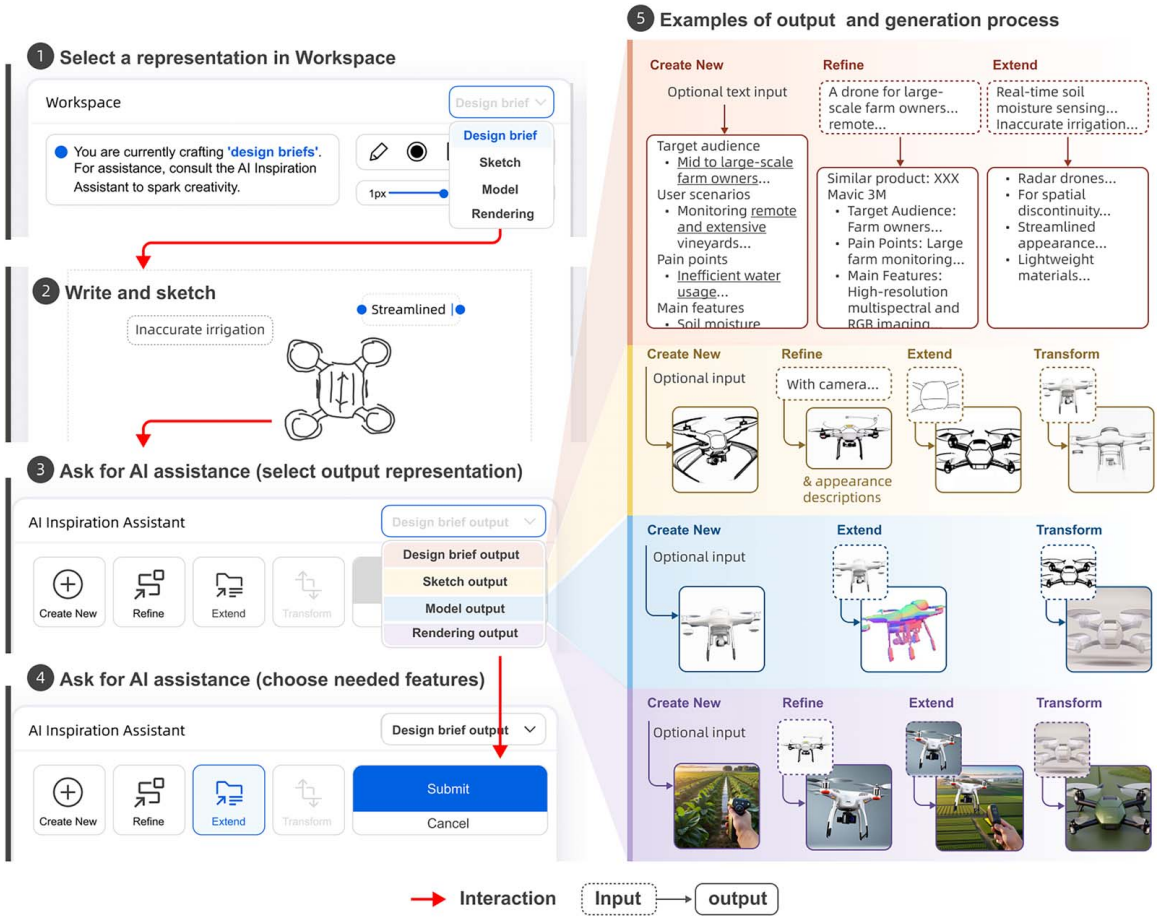


Fig. 6 StepIdeator provides features for four representation outputs in the AI Inspiration Assistant Panel to support interactions between the same and different representations. Designers can freely modify or transform any two representations. For each feature of every representation, we provide an example illustrating potential input and output scenarios.

For DG3, StepIdeator enables designers to tailor AI-generated outputs through preset prompts and personalized inputs, aligning closely with their conceptual visions. Details are shown in Sec. 4.3.1.

4.3 Implementation. StepIdeator is developed as a web-based platform, employing a REACTJS[12]-driven front-end and a FLASK[13]-powered back-end. It leverages ANT DESIGN[14] for styling and the "react-sketch-canvas"[15] component for freehand sketching capabilities. Textual content generation and design interpretation functionalities are facilitated through GPT-4V[16] via the OPENAI API. For image generation, it utilizes STABLE DIFFUSION V2.1 [73] and V1.5 using CONTROLNET [74], ensuring a wide range of visual creation possibilities. Additionally, the Segment Anything model [75] is used for segmenting an object from the background in images, and WONDER3D [76] is used to generate multiangle model images.

4.3.1 Crafting Prompts for Generating Different Design Representations. Prompts are user-provided textual instructions, denoting their design intent and controlling the generated content of AI models [77]. For each representation, targeted prompts are developed to ensure the quality and replicability of outcomes. First, we followed specific evaluation criteria based on prior research studies [77–80], including content relevance, diversity, output stability, and other criteria outlined in Appendix B, Evaluation Criteria for Prompt Iteration. Then two coauthors and one invited researcher specializing in generative AI collaborated on prompt iteration, experimenting with different wordings until there was a consensus that the outcomes met the evaluation criteria.

We use GPT-4V for text outputs due to its nuanced understanding of both textual and image inputs, along with its ability to generate relevant and creative content, making it well suited for complex design tasks. Based on guidelines from the prior research studies [81], our prompts are designed with a structured organization, including concept explanation, task brief, requirements, output criteria, and examples (Appendix B, Implementation Details of StepIdeator). Additionally, we break down the overall generation task into several simpler ones, organizing the targeted prompts sequentially for GPT to handle (detailed in Sec. 4.3.2).

For representations of image formats, we select STABLE DIFFUSION for its ability to customize image content through positive and negative prompts and ControlNet for its precision in image manipulation. The prompts for image generation consist of two parts: one part includes terms derived from iterative testing to standardize outputs into a specific representation, and the other part is generated by GPT, converting user inputs into prompts that meet STABLE DIFFUSION's generation requirements.

4.3.2 Implementing Features for Each Representation. Figure 7 depicts the technical details of StepIdeator. For each representation, tailored prompts and generative models are employed to effectively fulfill the corresponding features (Fig. 7(*b*)).

[12]https://react.dev/
[13]https://flask.palletsprojects.com/
[14]https://ant.design/
[15]https://github.com/vinothpandian/react-sketch-canvas
[16]https://openai.com/gpt-4

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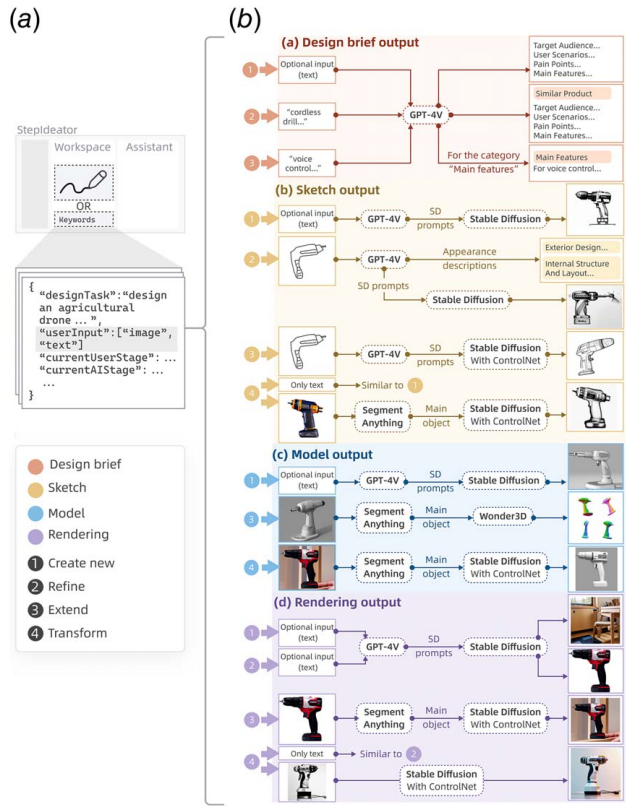


Fig. 7 Technical Overview of StepIdeator. To cater to designers' varying needs across different modalities, StepIdeator (*a*) captures text, canvas, or none inputs and (*b*) strategically combines several generative models, including GPT, STABLE DIFFUSION with ControlNet, the Segment Anything, and Wonder3D.

For outputs of design briefs (Fig. 7(*b.a*)), StepIdeator offers three features. The "Create New" feature leverages preset GPT prompts to generate content that aligns with the design task. The "Refine" feature utilizes the AI database to provide designers with potential similar products. Additionally, the "Extend" feature allows GPT to directly expand upon the designer's concept based on the provided prompts.

For outputs of sketches (Fig. 7(*b.b*)), the system provides four distinct features to facilitate the design process. The "Create New" feature uses specific combination prompts to activate STABLE DIFFUSION, producing relevant sketches that inspire designers with fresh creative insights. The "Refine" feature employs GPT-4V to provide appearance-based reference suggestions, which are then applied to the generation of STABLE DIFFUSION. To polish users' sketches, the "Extend" feature utilizes STABLE DIFFUSION with ControlNet's Scribbles model for generation. To "Transform" other representations into sketches, inputs are processed with ControlNet using the Lineart model. If the inputs are renderings with a complex background, the Segment Anything model is employed to extract the main object, which is subsequently transformed into a sketch using the ControlNet Lineart model.

For outputs of models (Fig. 7(*b.c*)), three features are supported. The generation logic for the "Create New" and "Transform" features is similar to that used for sketch outputs. When transforming sketches into models, the process utilizes STABLE DIFFUSION with ControlNet using the Softedge model. For transforming rendering inputs, the process first extracts the main object but then employs ControlNet using the Depth model for generation. Additionally, in the "Extend" feature, Wonder3D is invoked to expand single-perspective model images into multiangle views, thereby supporting designers in considering product structural details.

For outputs of renderings (Fig. 7(*b.d*)), all the four features are supported. The "Create New" feature uses STABLE DIFFUSION to create scene renderings without detailed products, preventing early fixation on specific solutions and enhancing immersion. Like other "Refine" features, the rendering version focuses on generating outputs based on similar products. The "Extend" feature allows users to extend model images or product renderings into scenes. We employ the Segment Anything model to extract the main object within user-input images and then use STABLE DIFFUSION to generate possible backgrounds. To generate detailed rendering images with the "Transform" feature, both sketch and model inputs are processed with ControlNet using the Lineart model.

5 User Study

To evaluate the effectiveness and usability of our system, we conducted a within-subject experiment with 16 participants in comparison with a baseline system. We also explored whether our system improves creativity and how it supports the creative process. This study aimed to answer the following research questions (RQs):

- *RQ1:* Compared with the baseline system, does StepIdeator support idea externalization and promote creativity through the flexible transition between different design representations?
- *RQ2:* During the transitions between design representations, how does StepIdeator influence the emergence of design inspiration sources?
- *RQ3:* How do designers and AI collaborate within StepIdeator to effectively transition between design representations?

5.1 Participants and Apparatus. We recruited 16 participants (P1–P16, age: $M = 23.86$, $SD = 1.15$) with over 3 years of experience in product or industrial design and involvement in more than three design projects. All are familiar with generative AI tools like CHATGPT, STABLE DIFFUSION, and MIDJOURNEY; 13 have used these tools in their projects. All participants reported having sufficient skills in sketching, modeling, and rendering to express their design ideas. To eliminate potential bias, the participants were not involved in the formative study or system development. The study lasted 2 h, and participants received monetary compensation after this study.

StepIdeator compares with the baseline system to evaluate its effectiveness in achieving design goals: enabling diverse input representations, precise control over these representations, and AI-enhanced incremental insights in design workflows. Thus, the baseline system has interfaces similar to those of StepIdeator but excluded encapsulated features such as "Create New" (Appendix C, Baseline System). The control group provided the same generative model as the experimental group contained in the system (e.g., GPT-4V, STABLE DIFFUSION with ControlNet, the Segment Anything model, and Wonder3D). Designers can upload images to the current canvas in the Workspace for editing or export the content from the canvas as an image and input it into the corresponding generative model to obtain the desired results.

During the experiments, both systems were accessed through the Chrome web browser on a 12.9-in. iPad equipped with a stylus, allowing participants intuitive interaction with the systems.

5.2 Study Procedure. Figure 8 illustrates the study procedure, consisting of two sections with StepIdeator and the baseline system separately. Participants were required to use two systems to complete different design tasks. The first task was to create a drone for monitoring agricultural parameters such as crop growth, soil moisture, and pest conditions. The second task involved designing an intelligent cordless drill specifically for users with disabilities, focusing on usability, accessibility, and safety. The order of tools and tasks was counterbalanced (detailed in Appendix C, Experiment Grouping).

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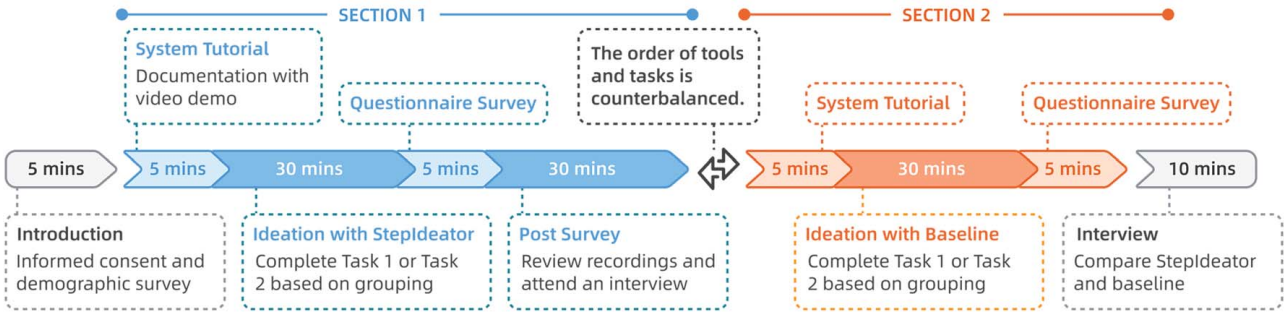


Fig. 8 User study process. The 2-h user study includes two sections with different tools, each with a 30-min ideation phase. The order of tools and tasks is counterbalanced.

To reveal the underlying patterns of inspiration within the mixed-design representation environment, in the section with StepIdeator, participants needed to review screen recordings of their design process, verbalize their thoughts, and receive follow-up interviews about the origins of their innovative ideas (Appendix C, Interview Questions). Participants were also required to explain all the inspiration and new ideas that occurred in their ideation process. The definition of inspiration usually references the influence of an external stimulus on the creative process, particularly affecting problem framing or solution generation [82–84]. In this article, we use the term "inspiration" to describe a creative shift in ideation regarding design solutions. The count of inspirations is determined by participants as they review video recordings, classifying each new operation or idea with guidance. After completing each ideation, participants were required to fill out a questionnaire (Appendix C, Self-Customized Questionnaires) and undergo a semistructured interview (Appendix C, Interview Questions).

5.3 Metrics. To validate StepIdeator's effectiveness in assisting idea externalization and enhancing creativity (RQ1), we focused on two aspects: (1) Idea Externalization System Usability Scale (SUS) [85] was applied to verify StepIdeator's usability in supporting idea expression. The SUS is a widely used questionnaire that comprehensively measures participants' subjective perceptions of the usability of a system, assessing efficiency, learnability, and other dimensions. Furthermore, a seven-point custom questionnaire (Appendix C, Self-Customized Questionnaires) was designed to assess designers' satisfaction with the output representations (Representation Satisfaction), adherence to initial concepts (Alignment), and satisfaction with the design outcomes (Outcome Satisfaction). (2) Creativity enhancement participants evaluated whether representation transition promoted creativity through creativity support index (CSI) [86]. The CSI assesses factors such as enjoyment, exploration, expressiveness, immersion, and results worth efforts to evaluate how well a tool supports creative processes. For the design quality, we invited three experts with 4–6 years of industrial design experience ($M = 25.67$, $SD = 0.58$) to rate the design outcomes. These experts had not participated in the formative study, user study, or system development. The expert rating metrics involved novelty and usefulness [87–89], and the computation details are provided in Appendix C, Calculation of Novelty and Usefulness.

To examine the effect of transitions between design representations on inspiration sources (RQ2), we derived patterns in these transitions via user interaction logs and timing records. Additionally, we recorded the timing and reasons for inspiration and new ideas when participants reviewed their design process and verbalized their thoughts. All combinations of representations within the mixed-design representation environment and the inspirations that occurred within these combinations were also tallied.

To capture participants' experiences with AI interaction (RQ3), we designed a self-customized questionnaire based on recent studies [52,90]. This questionnaire employs seven-point scales to evaluate designers' confidence, sense of ownership, contributions, and experiences of human–AI collaboration during representation transitions (Appendix C, Self-Customized Questionnaires).

For quantitative data, we initially assessed distribution normality using the Shapiro–Wilk test, followed by paired-sample *t*-tests for normally distributed data and Wilcoxon signed-rank tests for non-normal datasets.

6 Result

We report detailed quantitative and qualitative results from the user study, addressing three RQs.

6.1 Idea Externalization and Creativity Support

6.1.1 Idea Externalization. As indicated by SUS, StepIdeator ($M = 78.04$, $SD = 7.92$) demonstrated significantly greater usability ($p < .01$) in facilitating idea externalization compared to the baseline ($M = 61.25$, $SD = 9.66$).

Additionally, as shown in Fig. 9(*a*), participants using StepIdeator reported higher satisfaction with the output representations ($M = 6.06$, $SD = 0.68$). Their final designs received more favorable evaluations compared to the baseline in terms of adherence to initial concepts ($M = 5.56$, $SD = 1.26$), and satisfaction with design outcomes ($M = 5.81$, $SD = 0.83$). This demonstrated that our system effectively enabled designers to articulate their design ideas. P6 and P10 mentioned one possible reason for this, "Compared to the baseline, StepIdeator reduces frustration and cost associated

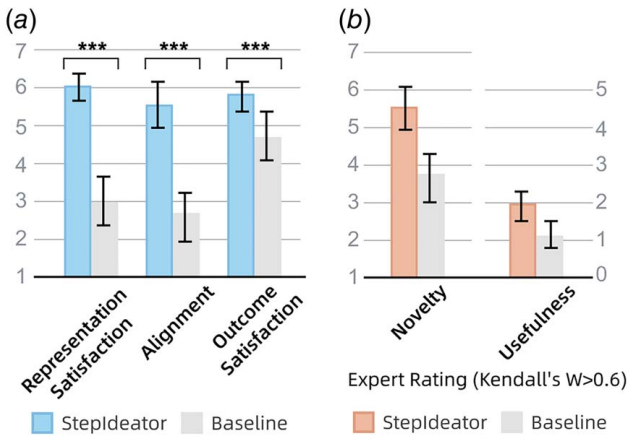


Fig. 9 Evaluation results of RQ1 with 95% confidence interval: (*a*) self-reported representation satisfaction, alignment, and outcome satisfaction for StepIdeator and the baseline and (*b*) expert-rated novelty and usefulness on design results for StepIdeator and the baseline system (*$p < 0.001$).**

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with each trial, increasing my willingness to iterate and modify designs, thus enhancing both the quality and quantity of output representations." P7 also emphasized that "StepIdeator effectively supports seamless transitions between representations, enabling deep exploration and presentation of design ideas from multiple perspectives. In contrast, the baseline cannot directly connect with my ideas, leading to wasted time and ineffective externalization."

6.1.2 Creativity Enhancement. Expert ratings of novelty and usefulness for StepIdeator and the baseline system demonstrated strong consistency, as evidenced by Kendall's W consistency test. For StepIdeator, Kendall's coefficients of concordance were 0.6 for novelty ($p < .05$) and 0.639 for usefulness ($p < .05$). For the baseline, the coefficients were 0.715 for novelty ($p < .05$) and 0.675 for usefulness ($p < .05$). The novelty ($M = 5.54$, $SD = 0.27$) and usefulness ($M = 1.97$, $SD = 0.19$) ratings demonstrated StepIdeator's ability to greatly inspire designers' creativity compared with the baseline system (novelty: $M = 3.73$, $SD = 0.30$; usefulness: $M = 1.12$, $SD = 1.67$) (Fig. 9(*b*)).

The CSI results also showed that StepIdeator ($M = 81.00$, $SD = 11.21$) significantly outperformed the baseline system ($M = 47.81$, $SD = 21.77$), especially in the dimensions of Exploration ($p < .01$), Expressiveness ($p < .01$), Enjoyment ($p < .01$), and Results Worth Effort ($p < .01$) (Fig. 10). In the Exploration dimension, all participants believed that StepIdeator helped them track ideas, outcomes, or possibilities. Ten participants mentioned that the various design representations provided by StepIdeator helped them expand their ideas. Regarding design space exploration, P1, P3, and P5 mentioned, "StepIdeator provides more suggestive content instead of just presenting results; it also provides intermediate steps of design, allowing designers to fully engage in divergent thinking." In contrast, half of the participants felt that the baseline system's interaction did not align with their design workflows, hindering their exploration process. For example, P2 and P15 pointed out, "The baseline outputs end-to-end renderings, limiting my conception and design space." P3, P6, and P7 mentioned, "The baseline's design process is AI-driven, fragmenting the design process, while conceptual design should be a continuous process."

In the Expressiveness dimension, all participants felt more creative when using StepIdeator. For example, 11 participants gained new inspiration during the process of transforming different design expressions: "During the transformation, AI may introduce new information, triggering new design inspiration and ideas; it can stimulate critical thinking, helping designers reflect on their schemes; comparing content before and after transformation also helps me associate it with other new ideas."

In the Enjoyment and Results Worth Effort dimension, participants "are willing to use this system frequently" and "are satisfied with its outputs." Regarding immersion, P1 also mentioned: "Operations that are smooth and aligned with design habits enable a focus on the design process itself."

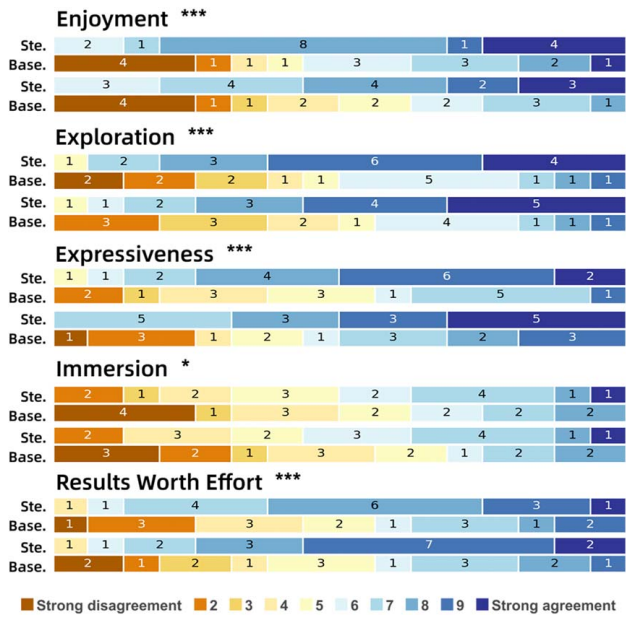


Fig. 10 Results of the CSI questionnaires. "Ste." and "Base." represent StepIdeator and the baseline system, respectively. *$p < 0.05$, ***$p < 0.001$.

6.2 Influence on Inspiration Sources. When using StepIdeator, 16 participants transformed or iterated design representations a total of 234 times, and 157 (67.1%) of these transformations inspired them. To understand the influence of mixed representations on inspiration, we (1) compared the combinations of design representations chosen by the participants and their rationales within the mixed-design representation environment and (2) analyzed sources of inspiration following each change.

6.2.1 Supporting Creativity Through Combining Mixed-Design Representations. Figure 11 visualizes the design process of all participants and their preferences for representations within the mixed-design representation environment when using StepIdeator. In Fig. 11(*a*), frequent interactions between participants and AI demonstrate how their collaboration iterates on design solutions through different design representations. First, participants varied in their reliance on representations, some favoring one or a few, while others distributed their usage more evenly. This variance in working styles underscores the need for CST to provide personalized content. Second, the participants were freed from the need to spend extensive periods guiding AI to regenerate outputs, devoting more time to conceptualizing and iterating designs. In Fig. 11(*b*), we calculated the proportion of time spent by all the participants on creating a certain type of representation with the assistance of different representations from AI. First, for each type of representation, participants not only selected representations that matched their ongoing creation but also tended to request more concrete representations in the AI Assistant panel when working on different types of representations. Second, when comparing different representations, participants spent more time on the design briefs and sketches, while relatively less time was devoted to models and renderings.

Similarly, as evidenced by Fig. 12, designers often opted for more concrete and vivid AI representations for reference in a mixed-design representation environment from StepIdeator. For example, they requested AI to provide model images as references while sketching. Participants also reported experiencing an "attention shift" in a mixed-representation environment [12] when they maintained the same representation for their conceptualization but switched to different representations in the AI panel. They described this shift as a means to derive "more abstract or more concrete inspiration stimuli." Transitioning between representations, StepIdeator helped participants "obtain new externalized forms based on the previous design content, preserving thought coherence and fostering innovation."

Specifically, we noted that 15 participants started conceptualizing from design briefs (Fig. 11(*a*)), which provided the most inspiration according to their accounts of the design process (Fig. 12). During interviews, ten participants expressed that "text is suitable for inspiring ideas because it contains enough information." When writing design briefs in the Workspace, designers often requested AI assistance to generate or refine their briefs (17.1% in 234 times of transformations), display relevant sketches for references, or transform design briefs into sketches (14.7%). Scene renderings (3.4%) and model images (2.6%) were also considered helpful to enhance immersion. When designers concentrated on the design brief, they could "supplement background knowledge" and "determine the embryonic form of the design." When they simultaneously

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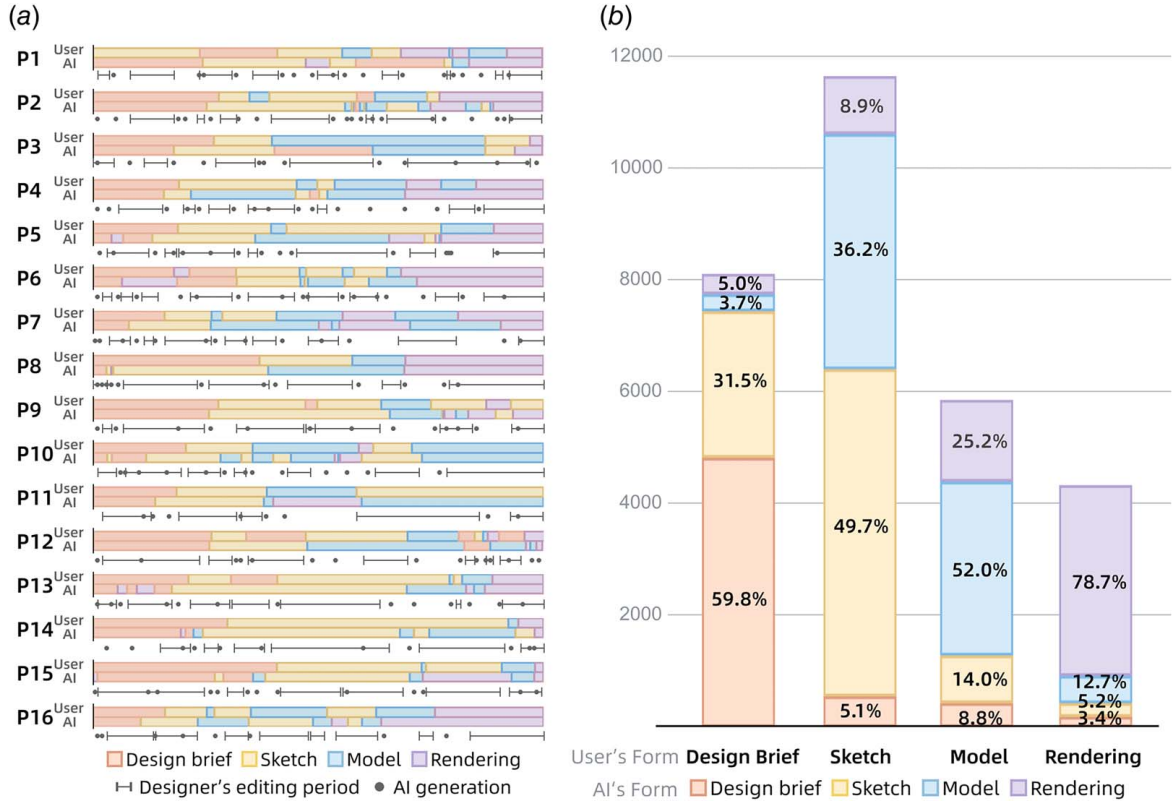


Fig. 11 Visualization chart of the comparison of time spent on design representations between participants and AI in StepIdeator. (*a*) The evolution of design representations for all participants, with each record including both the designer's and the AI's contributions. To describe the design process of each designer when using StepIdeator, the lines represent the designer's editing periods, while the dots represent the actions of AI generation. (*b*) The distribution of the total time (seconds) spent by all designers on creating a certain type of representation with the assistance of different representations from AI.

wrote the design brief and browsed AI-generated sketches, multiple participants verified "whether the defined functions are reasonable from sketches" or allowed AI to "organize logical concepts from scattered text and transform them into visual appearances."

Additionally, sketches are the most frequently used form of representation in the design environment (50.7% occurrence), and they account for the longest duration of work (38.71% for designers' time, 31.3% for AI generation time). This may reflect the central role of sketching in idea generation and initial visualization.

While sketching, designers were "exploring different design directions or need more help iterating their ideas quickly." For this representation, designers and AI showed the most diverse collaboration patterns within the mixed-design representation environment. Activities included designers sketching alongside AI-generated model images (11.5% in 234 times of transformations), requesting AI to refine sketches (15.8%), and integrating sketches with rendering (4.3%) and design brief outputs (1.3%). When drawing sketches and simultaneously requesting AI-generated model images, eight participants mentioned that "StepIdeator allows me to iterate between sketches and models, which helps me verify the form and structure by examining it from multiple perspectives."

As participants completed the two high-fidelity representations, models, and renderings, we saw that AI-supported representations largely aligned with representations in Workspace. These two representations also used less AI time overall, with P1 and P12 noting that they "relied more on their own expertise and did not need much creative input."

6.2.2 Inspiration After Each Change in Artificial Intelligence-Provided Design Representations. After a designer uses any feature in the AI Inspiration Assistant Panel, AI generates new content, refines ideas, extends design schemes, or transforms into another representation. Observations suggest that 157 instances of inspiration arise through this process with StepIdeator (67.1%). To analyze the patterns of innovation when participants designed with mixed representations using StepIdeator, we analyzed the sources of inspiration after each change in AI-provided design representations and identified three sources of inspiration (Fig. 13).

Inspired by New Information From AI (57 Instances): During the process of transforming design representations, new data, perspectives, or insights may be introduced into the design process. This addition not only "expands the existing design information base", but may also "reveal new possibilities and directions." P5 and P10 further mentioned: "Some transformations stimulate associative thinking." For example, while using StepIdeator for an agricultural drone design, P2 drew inspiration from the AI-generated sketch based on her design brief's keywords "precise" and "small-dose delivery of growth hormones." The shape of the drone depicted in the sketch sparked the idea: "This shape is very regular. Perhaps I can use a similar shape to design a drone array. Each drone could carry a payload of chemicals underneath."

Reflecting on Ideas Provided by AI (63 Instances): Designers scrutinize AI-generated content to refine their design proposals, often leading to novel inspiration. This critical examination helps distill core design issues and isolate viable ideas by cutting through extraneous details. "The refinement process aids in clarifying desires and resolving potential issues, minimizing the need for persistent trial and error," as noted by participants. For instance, P9 expressed unfamiliarity with the product and pondered improvements to the electric drill's accessibility, contemplating a smart-operated version. Yet, when the system proposed "…an intelligent control panel to support simple operations," P9 clarified, "It

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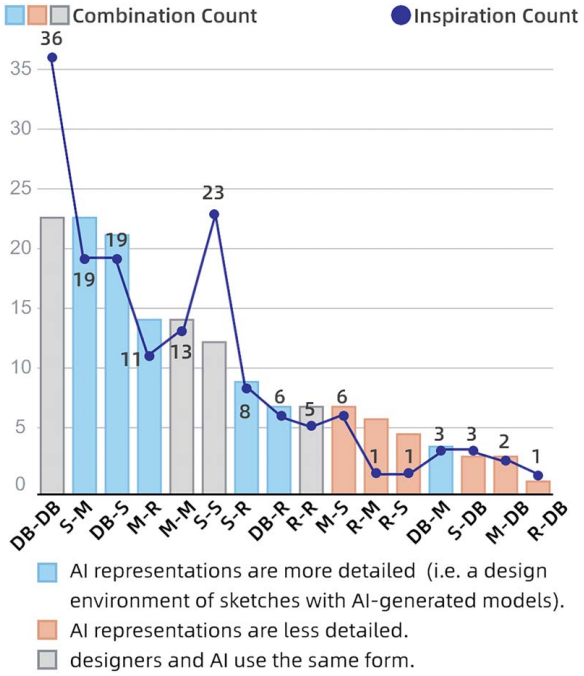


Fig. 12 Count of each design representation combination and inspiration in StepIdeator. Representations include design briefs (DB), sketches (S), models (M), and renderings (R). The former abbreviation represents the designer's representation, and the latter represents AI's representation (i.e., "DB-S").

can't be that kind of smart. The operation should be straightforward; perhaps we could design mechanical buttons."

Inspired by Representation Transitions (68 Instances): In the process of transforming design representations, designers map the same scheme or concept into multiple representations. This transformation allows designers to "examine problems from different perspectives and dimensions in a short time," "facilitating a deeper understanding of design challenges," and "potentially revealing solutions not previously considered." For example, after drafting a simple sketch of a drone, P4 transitioned it to a model image, which sparked new inspiration. P4 noted, "This top view is precisely what I was seeking. Yet, it prompts a reconsideration of the wing and support positioning. I'll adjust my sketch accordingly."

These sources do not exist independently; a particular inspiration involves combinations. Furthermore, both P1 and P11 described the process of using StepIdeator for the product concept design as an open iteration: "This process involves incremental information, filtering, mapping, and AI continuously introduces new content to prevent thinking from solidifying." P7 added, "I need to constantly add or subtract, splice, and assemble the transformed content to finally output my conceived solution."

6.3 Understanding Human–Artificial Intelligence Collaboration Experience

6.3.1 Confidence in Artificial Intelligence-Generated Representations. When using StepIdeator, participants generally reported enhanced confidence in the AI-generated content (Table 1). P8 mentioned that "Refining the content incrementally with StepIdeator boosts my confidence." Participants also demonstrated significant variations in their trust levels toward AI-generated representation when externalizing their design concepts.

6.3.2 Sense of Ownership and Contribution. In the interviews with StepIdeator, eight participants expressed a sense of ownership over their design, and all participants firmly believed that they had the ultimate right to interpret their designs. Participants noted that the final output reflected their deep thinking about the design task. Although AI offered prompts and suggestions, the primary conceptualization was driven by humans. StepIdeator delivered initial suggestive content to foster autonomous idea development and facilitated precise adjustments to details to meet expectations. Conversely, the baseline system seemed more dominated by AI. P7 mentioned, "When using the baseline system, I felt that my energy was primarily spent on controlling the visual effects of the images, gradually convincing myself to accept the outputs it provided after repeated attempts. In contrast, when using StepIdeator, I felt in control of the design process, conceiving a product from scratch as a designer." P1, P2, and P5 also expressed similar sentiments.

6.3.3 Experience of Collaborating With Artificial Intelligence. Participants using StepIdeator scored higher on their experience of human–artificial intelligence collaboration ($M = 6.06$, $SD = 1.34$) than the baseline ($M = 5.19$, $SD = 1.01$) with a p-value of 0.001. "Designing with StepIdeator is more like working with a design assistant that understands the product design process and conforms to design habits," as mentioned by P1 and P5. P7 looked at the possible future of StepIdeator: "AI's ability to transform representations can compensate for possible skill shortcomings in a designer's design process and improve design efficiency."

7 Discussion

We introduce StepIdeator, a generative AI-powered design tool that enables a fluid progression and enhancement of various design representations. Drawing from our observations, we recommend several design considerations for the next generation of AI-based creativity support tools. We also discuss the limitations and future work of this study.

Challenges and Opportunities of Enhancing Idea Externalization: The results show that our step-by-step approach facilitates idea externalization in AI-supported conceptual design, where transformations among various design representations streamline the conceptual design process. Nevertheless, the effectiveness of idea externalization requires further improvement due to the ambiguity of designers' ideas [7] and the uncertainty and opacity of AI [72]. For instance, P3, P4, and P12 noted that "directly generating models from sketches makes it difficult to correct certain details of the model" when utilizing the "Transform" feature for sketch-to-model generation. P4 further explained, "I might identify structural issues in my initial sketch upon viewing the AI-generated model image. However, this creates a dilemma: accurately depicting the correct structure in my rapid hand-drawn sketches is challenging, and allowing AI to directly modify the model image could result in the creation of unwanted elements."

The dilemma illustrates one of the biggest challenges with current generative AI methods: editability control. In the process of generating one type of media from another, researchers have tried to address the issue of editability control through methods such as enhancing the granularity of prompt control [52,91], imposing constraints from a third type of media on the generation process [16,17], and utilizing iterative feedback loops [92]. These approaches tackle editability from three aspects, respectively: the generation command, the generation process, and postgeneration editing. Our research is a synthesis of the second and third approaches. We propose a method that breaks down the AI generation process step by step, aiming to segment its generation into editable and iterative processes that align with the design process. Yet our system lacks support for finer-grained control over detailed areas with prompts, such as partial sketch structures. Therefore, we believe that future control of generative AI may further integrate multiple dimensions for a comprehensive approach.

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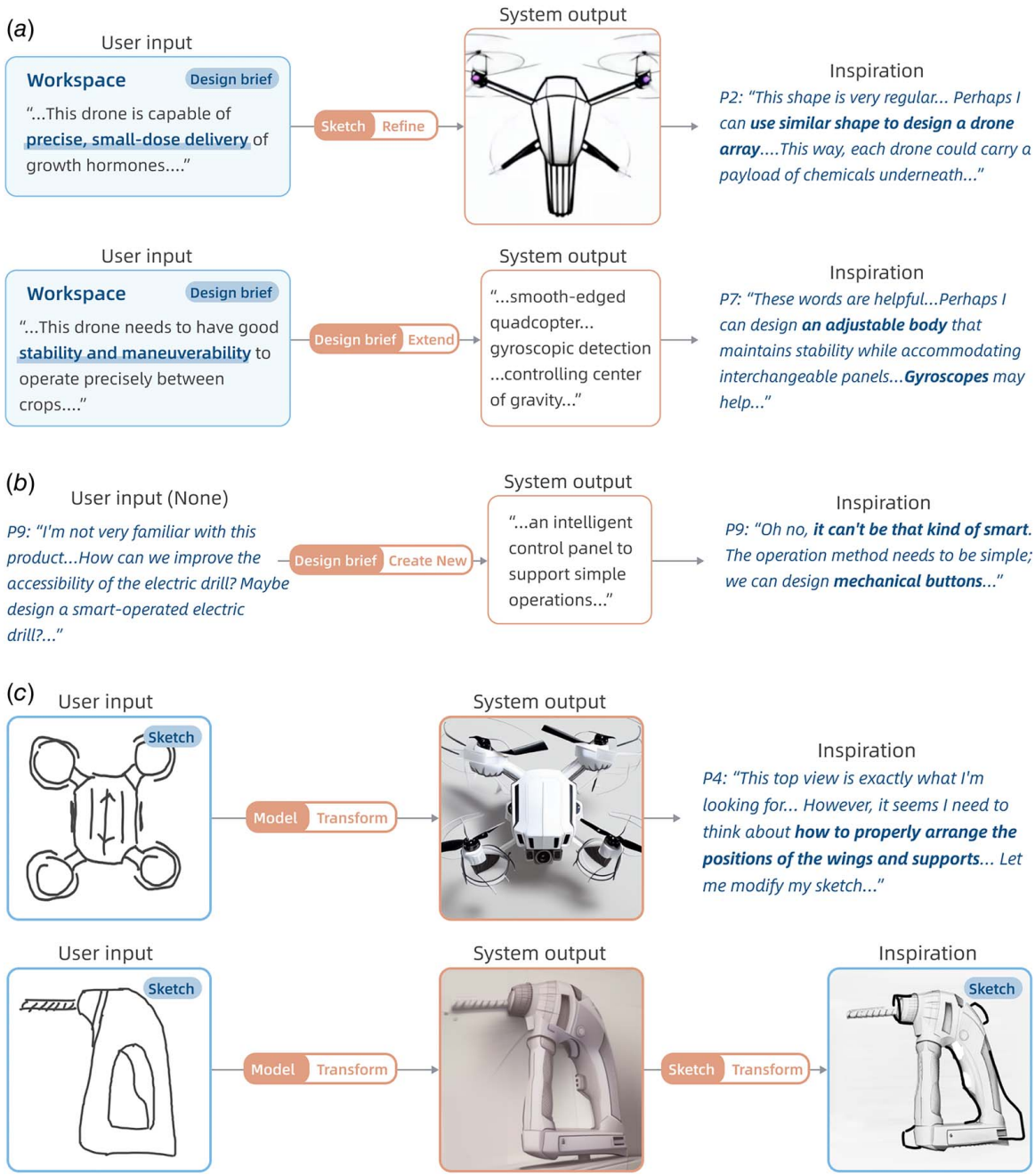


Fig. 13 Inspiration sources of designers. Designers gained insights and innovations from (*a*) previously unconsidered information from AI, (*b*) reflective scrutiny of AI suggestions, and (*c*) transitions between different representations, enriching their design processes.

Two Approaches to Future Design With AI: Speed and Volume versus Unique Creativity: In our user study, participants expressed high satisfaction with StepIdeator and are optimistic about the potential of similar tools in future design practices, suggesting this approach will influence design methodologies. The rapid advancement of AI has significantly influenced various industries, necessitating adjustments in design processes to balance productivity and creativity [20]. In practice, design tasks fall into two categories: those prioritizing speed and volume, where innovation is secondary, aiming for the quick generation of multiple solutions [93]; and those emphasizing unique creative outputs, where innovation surpasses speed [94]. For the former, AI facilitates rapid transitions from initial concepts to detailed visual effects. For tasks focused on innovation, while AI inherently does not follow a step-by-step process, it must adapt to the designer's need for structured, progressive methods, fostering subtle transitions from broad concepts to detailed articulations. For example, P16 pointed out, "StepIdeator is still not efficient enough, particularly in the information quantity of image outputs. It cannot provide a reasoning process for the shaping in sketches." Thus future research could build on our findings, exploring how the use of AI is balanced with the designer's ideation process.

Two Roles of AI in Future Human–AI Collaboration: The results indicate that participants using StepIdeator feel a greater sense of ownership. This effect might stem from the role of AI within the system. P1 and P5 noted that while AI functions more as an "executor or independent creator" in the baseline, AI acts as a "collaborative designer" in StepIdeator. Furthermore, the sense of ownership may also derive from the distinct nature of ownership in design versus art. The conceptual design prioritizes core

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Table 1 Results of designers' confidence in AI-generated content during ideation

Designer	AI	StepIdeator	Baseline	Sig.
Design brief	Design brief	6.12	5.37	–
Design brief	Sketch	5.00	4.00	*
Design brief	Model	5.13	3.68	*
Design brief	Rendering	4.81	4.18	–
Sketch	Design brief	5.68	5.00	–
Sketch	Sketch	5.19	4.06	*
Sketch	Model	4.88	4.31	–
Sketch	Rendering	5.06	4.18	–
Model	Design brief	5.75	4.93	–
Model	Sketch	5.69	3.88	**
Model	Model	5.06	4.25	–
Model	Rendering	5.37	4.31	*
Rendering	Design brief	5.75	4.81	*
Rendering	Sketch	5.50	3.68	***
Rendering	Model	5.06	4.18	–
Rendering	Rendering	5.13	4.13	*

Note:
–$p > 0.05$.
*$p < 0.050$.
**$p < 0.010$.
***$p < 0.001$.

functionalities over aesthetics [19]; thus, designers' outputs are comprehensive solutions that integrate functionality, structure, and appearance. Our step-by-step approach further enhances designer involvement in functional aspects, reinforcing their ownership.

Limitations and Future Work: First, designers using StepIdeator were limited to toggling between only two parallel representations, and we did not explore conditions involving more than two forms. One participant commented during the experiment, "I wish I could see all four forms directly in the workspace." This could be a direction for future research.

In StepIdeator, text and images were used to encapsulate the roles of different design representations in the conceptual design. However, designers often require editable 3D models instead of merely model images when transitioning from conceptual design to production. Although we tried to fully utilize the current capabilities of AI to provide multiangle model images, discrepancies still exist between these forms and the models used in practical scenarios. The advancement of 3D generative technologies is expected to address these issues in the near future.

Furthermore, as both StepIdeator and the baseline employed generative AI techniques including GPT and STABLE DIFFUSION, the results may be influenced by users' familiarity with AI. While our participants were highly knowledgeable about AI, we cannot predict how other designers may perceive such tools. This suggests a potential avenue for future research to explore how varying levels of familiarity with AI affect the creativity of designers using AI-driven tools.

8 Conclusions

In this article, we explored how generative AI supports designers in externalizing concepts with progressive design representations. In a formative study, we identified three key challenges in integrating AI into the design thinking externalization process. Then we developed StepIdeator, an AI-based system capable of iterating and transforming between mixed representations including design briefs, sketches, model images, and renderings. A user study involving 16 participants provided empirical evidence of StepIdeator's ability to enhance deep engagement in the design process, creativity, and innovation. Participants reported a heightened sense of ownership and confidence, indicating a substantial improvement in the human–AI collaboration experience. On the basis of the findings, we draw insights into how we should design and build future generative tools.

Funding Data

- The National Key R&D Program of China (2022YFB3303301).

Conflict of Interest

There are no conflicts of interest.

Data Availability Statement

The authors attest that all data for this study are included in the paper.

Appendix A: Interview Questions of Formative Study

Designers were asked the following questions at the beginning of the study:

(1) Could you describe your typical design process? For example, how would you organize the entire design process if you received a design request for an "intelligent vacuum cleaner"? What steps would you divide it into?
(2) What representations do you generally use during the design process? Why do you choose this form? What tools or software do you use when employing this representation? What are the advantages and disadvantages of this representation?
(3) Do you use AI tools during the product design process? In which stages or steps of the design process do you typically use them? What is the purpose of using these tools? Have you encountered any problems during the generation process with these tools?

After the generation tasks, we introduced the purpose of the study and our preliminary solutions: "To assist designers in flexibly producing various design representations and switching between them seamlessly, we aim to design an AI-assisted system. Users can switch between different types of representation environments as needed. During the design process, users have access to a freehand drawing main canvas, along with an AI assistant panel on the right…" Then designers were asked the following questions in the postsurvey interviews:

(1) What problems did you encounter during the generation process with these tools?
(2) During the design process, in which types of representation environments would you prefer to switch? Why?
(3) Compared to traditional sequential switching in the design process, what are the advantages and disadvantages of this method?
(4) At which points in the conceptual design process do you feel AI assistance is most needed?
(5) What role do you expect AI to play when you are concretizing a certain design representation?
(6) Do you think you would be willing to use this system? Why?
(7) How many outputs should the system generate at once? Should a brief textual description accompany the generated images? Is it necessary to clearly distinguish between different design representation environments?
(8) Would you prefer to switch environments sequentially or be able to jump freely between them?

Appendix B: Details of StepIdeator

Evaluation Criteria for Prompt Iteration

(1) Instruction Compliance: Whether the generated stimuli follow the instruction format.
(2) Content Relevance: Whether the generated stimuli are relevant to the input task or design ideas.
(3) Output Stability: Whether consistent quality results can be produced when executed ten times.

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Fig. 14 Example of prompts for the "Create New" feature of design briefs

(4) Definition Compliance: Whether the outputs meet the definition of design representations.
(5) Diversity: Whether the concepts of different outputs are non-repetitive, are diverse, and can guide designers to think in different directions.
(6) Rationality: Whether the outputs are reasonable in design.

Implementation Details of StepIdeator. Figure 14 shows an example of prompts for the "Create New" feature of design briefs. Based on the past research and iterative results [81], the final structure of prompts includes concept explanation, task brief, requirements, output criteria, and examples. We have also uploaded the project on GitHub.[17]

Appendix C: Details of User Study

Baseline System. Figure 15 shows the interface of the baseline system used for the user study. Overall, the baseline system has interfaces similar to those of StepIdeator but does not offer the features in the AI Assistance Panel specifically designed for design representations. Instead, it provides the original models for designers to use. Designers can upload images to the current canvas in the Workspace or export a specified canvas for generation purposes. Additionally, designers have access to GPT-4V, STABLE DIFFUSION, and other generative models on a separate screen, with all models being identical to those in StepIdeator.

Experiment Grouping. Table 2 shows the experiment grouping of the user study. Participants were evenly distributed among four groups, each comprising four individuals. The distribution ensured equivalence across groups in terms of design experience and familiarity with AI tools. In the experiment, each group of participants sequentially completed two sections. We arranged the design tasks and systems in counterbalanced combinations to mitigate the potential impact of order effects.

Interview Questions. Participants were asked the following questions while reviewing video recordings of their experiment sessions with StepIdeator:

(1) Upon completion of the questionnaire, what were the reasons for the assigned ratings?
(2) What were the motivations and reasons for altering design representations? In what ways did the AI-provided design representations assist in this process?
(3) In each instance of feature use, how did the system provide assistance? Were the outcomes in line with expectations? What impact did this assistance have on the creation of design representations?
(4) What were the reasons for not utilizing certain features? What perceived impact might these unused features have had on the design representations?
(5) Upon generating new ideas, what were the underlying reasons for these developments? Please categorize these ideas.
(6) How did the design concept evolve throughout the experimentation process? Indicate specific time points of notable changes.
(7) What differences were observed between using StepIdeator for conceptual design and the usual process of conceptual design?

Participants were asked the following questions after completing tasks using the baseline system:

(1) Upon completion of the questionnaire, what were the reasons for the assigned ratings?
(2) How were the novelty and practicality of the design assessed?

At the end of the experiment, participants were asked to:

(1) Compare the advantages and disadvantages of the two systems.
(2) Elaborate on the differences between StepIdeator and the baseline when you worked on design briefs, sketches, model diagrams, and renderings.
(3) Describe the thought process when transitioning between different design representations and how new ideas were formulated or existing designs advanced.

Self-Customized Questionnaires. A seven-point custom questionnaire for idea externalization (RQ1):

(1) I am satisfied with my design representation outputs.
(2) My final design is consistent with my initial concept.
(3) I am satisfied with my final design.

A seven-point custom questionnaire for human–AI collaboration (RQ3):

Confidence in AI-Generated Representations

(1) How much trust do you place in the AI-generated design brief/sketch/model/rendering when creating the design brief?
(2) How much trust do you place in the AI-generated design brief/sketch/model/rendering when creating the sketch?
(3) How much trust do you place in the AI-generated design brief/sketch/model/rendering when creating the model?
(4) How much trust do you place in the AI-generated design brief/sketch/model/rendering when creating the rendering?

[17] https://github.com/Hualeez/StepIdeator/tree/master

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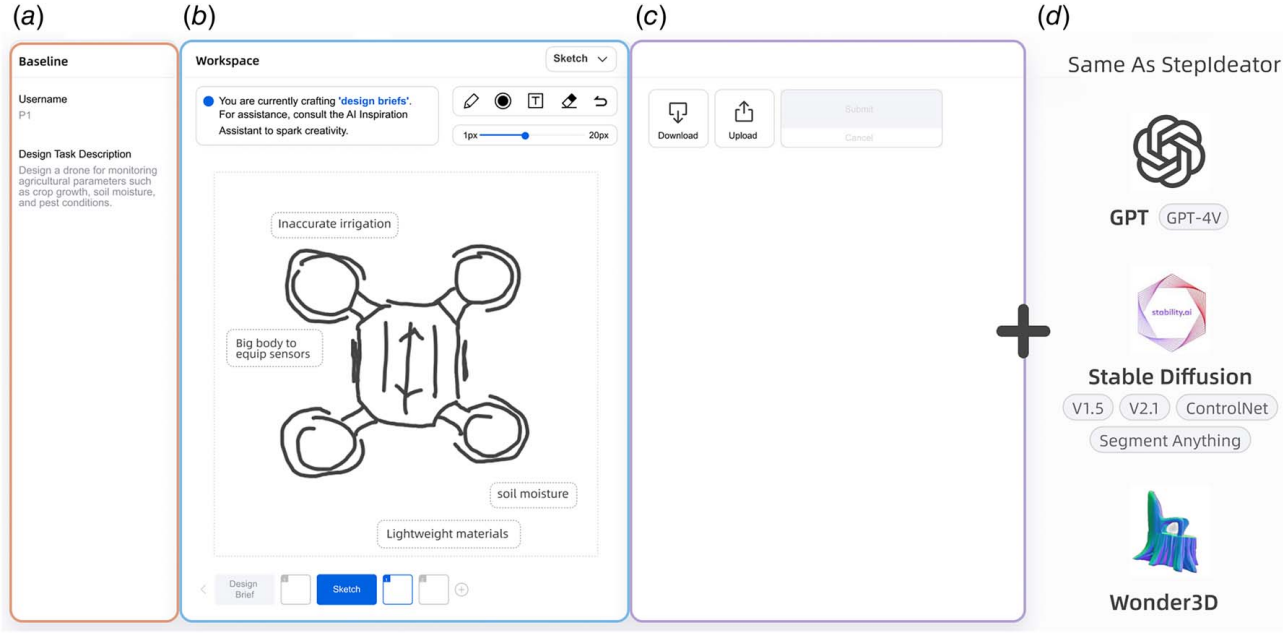


Fig. 15 Screenshot of the baseline. Similar to StepIdeator, the baseline also has three panels: (*a*) a Basic Information Panel, (*b*) a Sketch Canvas (Workspace), and (*c*) a Canvas Function Panel. In the Canvas Function Panel, designers can upload images to the current canvas in the Workspace or export a specified canvas for generation purposes. (*d*) Designers are also provided access to GPT-4V, STABLE DIFFUSION, and other generative models on a separate screen, with all models being the same as those in StepIdeator.

Table 2 Counterbalanced order of experiment grouping

	Section 1	Section 2
Group 1	Task 1 with StepIdeator	Task 2 with the baseline
Group 2	Task 2 with the baseline	Task 1 with StepIdeator
Group 3	Task 2 with StepIdeator	Task 1 with the baseline
Group 4	Task 1 with the baseline	Task 2 with StepIdeator

Sense of Ownership and Contribution

(1) How significant is your contribution to the design when creating the design brief?
(2) How much control do you have over the final decisions of the design plan when collaborating with AI during the design brief phase?
(3) How significant is AI's contribution to the design when creating the design brief?
(4) How significant is your contribution to the design when creating the sketch?
(5) How much control do you have over the final decisions of the design plan when collaborating with AI during the sketch phase?
(6) How significant is AI's contribution to the design when creating the sketch?
(7) How significant is your contribution to the design when creating the model?
(8) How much control do you have over the final decisions of the design plan when collaborating with AI during the model phase?
(9) How significant is AI's contribution to the design when creating the model?
(10) How significant is your contribution to the design when creating the rendering for products and scenes?
(11) How much control do you have over the final decisions of the design plan when collaborating with AI during the rendering phase?
(12) How significant is AI's contribution to the design when creating the rendering for products and scenes?

Experience of Collaborating With Artificial Intelligence

(1) How would you describe your experience collaborating with AI when creating the design brief?
(2) How would you describe your experience collaborating with AI when creating the sketch?
(3) How would you describe your experience collaborating with AI when creating the model?
(4) How would you describe your experience collaborating with AI when creating the rendering for products and scenes?
(5) As a designer, you expect AI to assist you as a design partner in: a. Discovering new design inspiration and trends. b. Assisting with design visualization and presentation. c. Creating innovative design solutions. d. Supporting the testing and evaluation of design effectiveness. e. Aiding in the training of professional skills and knowledge. f. Facilitating effective communication and mediation among different stakeholders.

Calculation of Novelty and Usefulness. Experts rated the overall solutions using a seven-point scale for novelty (N). Usefulness (U) is calculated as the product of the level of importance (L), the popularity of product usage (R), and its frequency (F). L is assessed using a five-point scale, while R and F are measured on a scale from 0 to 1, precise to one decimal place. $U = L \times R \times F$. We performed Kendall's W consistency test on the scores for N and U.

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